

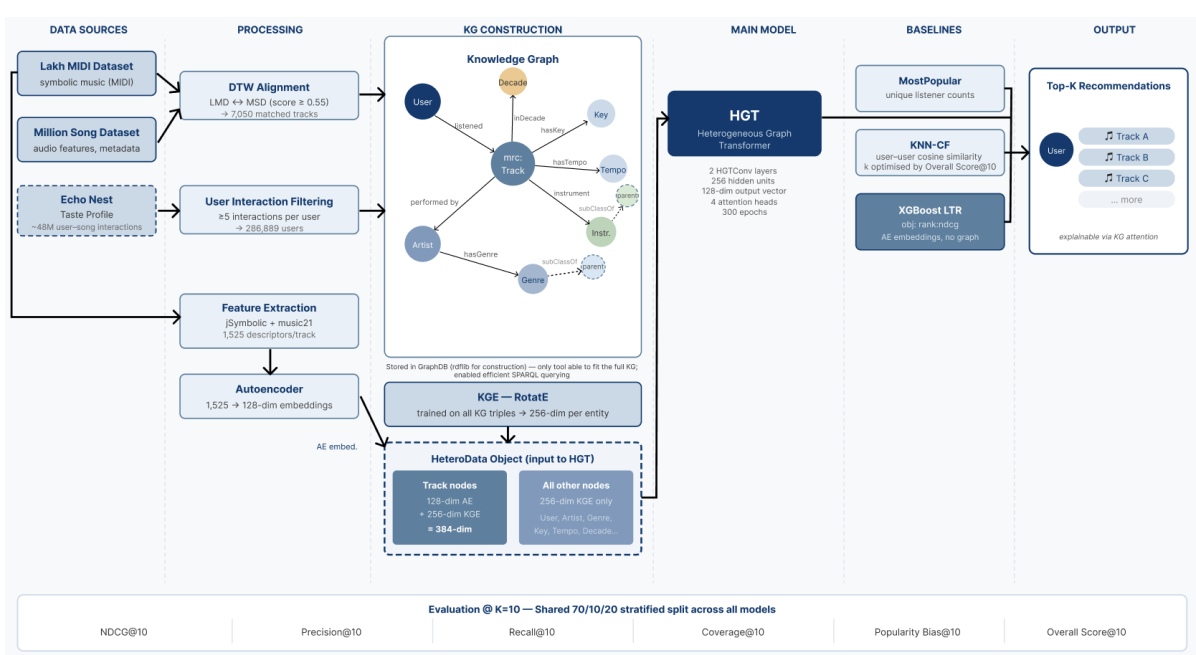
Objectives

- Recommend new music by enriching collaborative filtering (CF) with **symbolic + audio content** — genre, artist links, instrumentation, tempo.
- Encode it as a typed **Knowledge Graph** relating users, tracks, artists and genres beyond a user-item matrix.
- Learn over the KG with a **Heterogeneous Graph Transformer (HGT)** [1] for diversity, cold-start robustness and **explainability**.

Problem

- CF uses only interaction history [11] → **popularity bias** and **cold-start** failures, ignoring musical content.
- Most KG-enhanced work uses shallow embeddings [8, 10], without a GNN that propagates **multi-hop typed** structure [9].
- Can symbolic features + a heterogeneous KG + an HGT reason over shared genre / artist / instrumentation to close that gap?

Methodology



HGT [1] message-passing over the undirected KG; ℓ_2 -normalised embeddings → cosine scoring. Trained with a **custom listwise loss** (not pointwise binary cross-entropy nor pairwise BPR [11]): a soft cross-entropy over each user's candidate softmax, logit-adjusted to down-weight popular tracks:

$$s_i = \frac{\mathbf{u}_i^T \mathbf{t}_i}{\tau} + \lambda \log p_i, \quad \mathcal{L} = -\frac{1}{|B|} \sum_i \tilde{p}_{ui} \log \text{softmax}(s_{ui})$$

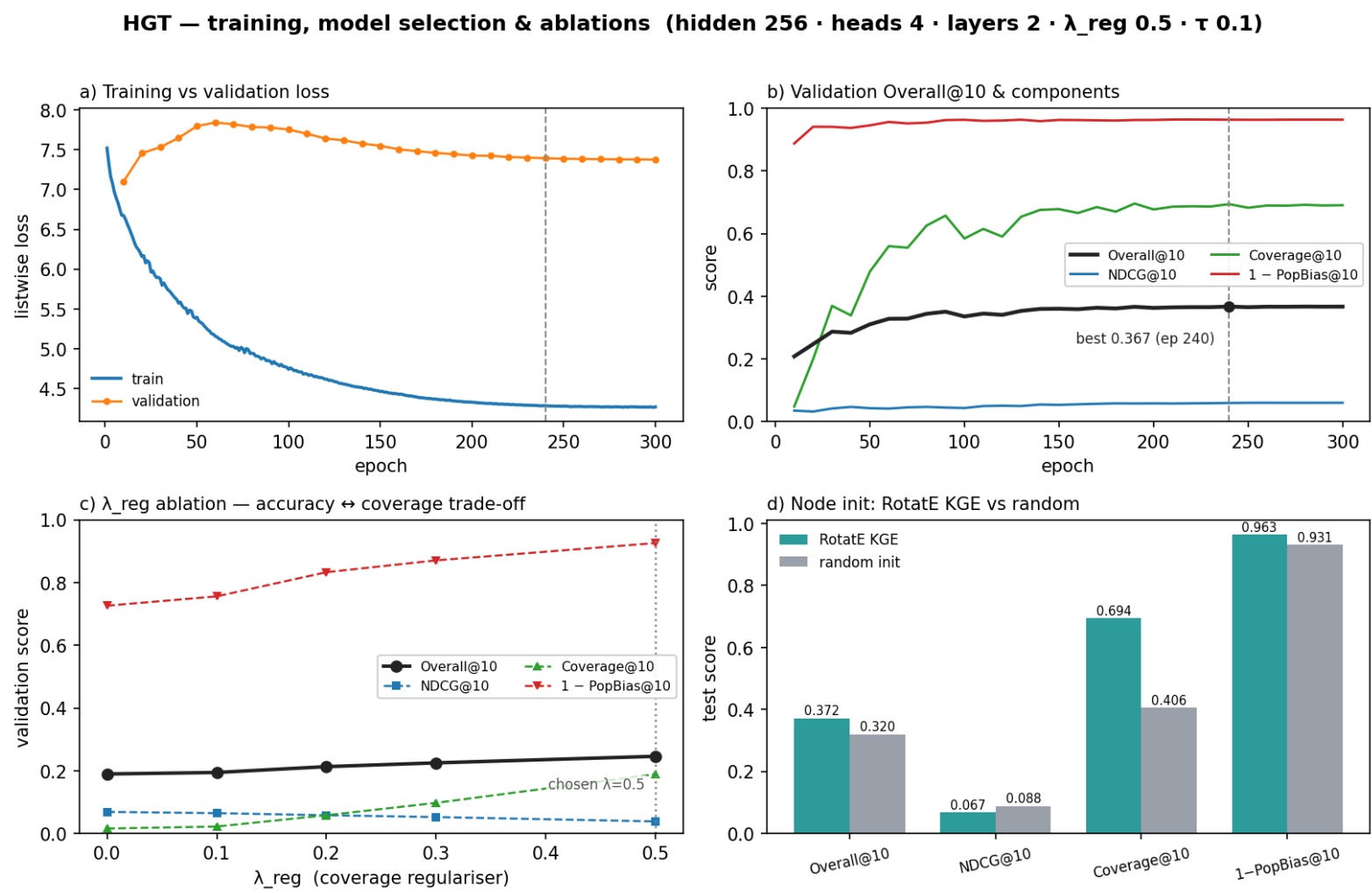
$\tau=0.1$; soft targets $\tilde{p}_{ui} \propto \log(1+c_{ui})$. As $\log p_i < 0$, the $\lambda \log p_i$ term **boosts less-heard tracks** — driving the low popularity bias below. Models selected by **Overall Score@10**= 0.6 NDCG+0.2 Cov+0.2(1-PB) — relevance, diversity & fairness.

Baselines

each: features · preprocessing · objective → output

- MostPopular** — global play counts, no user features; rank by frequency → one list shared by all users.
- KNN-CF** — binarised, ℓ_2 -normalised interactions; user-user cosine kNN ($k=6$) → neighbour-weighted scores.
- XGBoost-Hybrid** — AE user+track embeddings + 44 categorical (1-hot key/mode/tempo; target-enc. genre/artist; multi-hot instrument); 200 AE-cosine candidates/user, gradient-boosted LTR re-rank (**rank:ndcg@10**) → top- K .

HGT Training & Ablations



(a) train/val listwise loss; (b) validation Overall@10 & components; (c) λ_{reg} accuracy-coverage trade-off; (d) RotatE-KGE vs random node init.

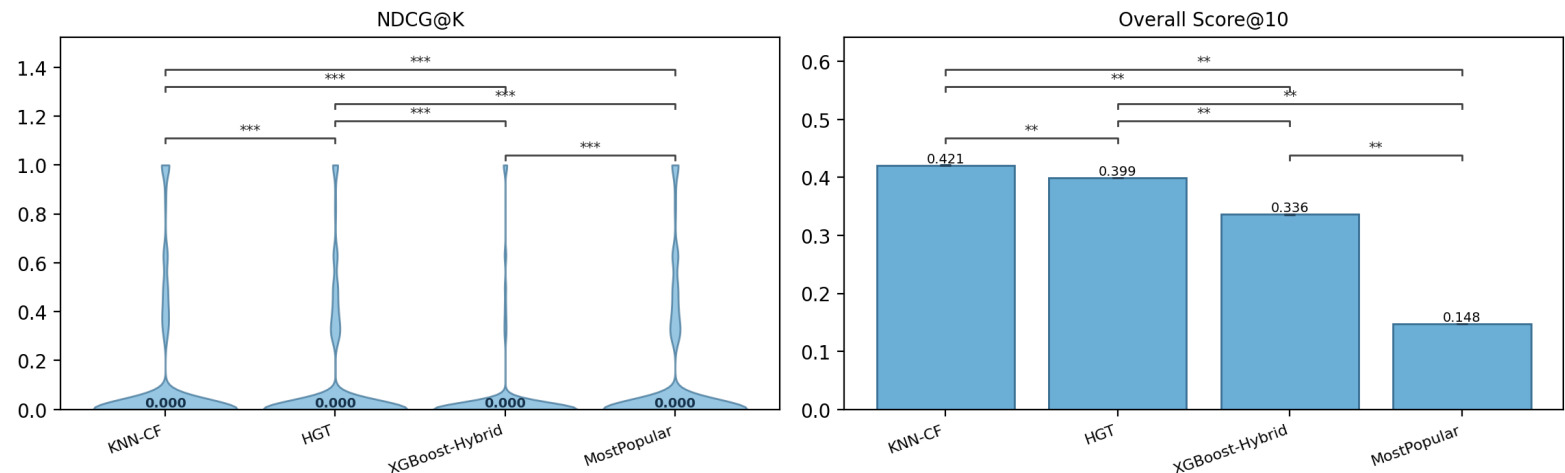
- Best config: **hidden 256, 4 heads, 2 layers**, $\lambda_{\text{reg}}=0.5$, $\tau=0.1$, early-stopped at epoch ~240 on validation Overall@10.
- RotatE-KGE node init** [2] clearly beats random (Overall@10 0.373 vs 0.329): the pretrained KG embeddings carry useful structure.

Recommendation Benchmark

Benchmark — NDCG & Overall Score (model × K)

	NDCG						Overall Score				
	5	10	20	50	100		5	10	20	50	100
HGT	0.079	0.097	0.113	0.133	0.148		0.372	0.399	0.425	0.456	0.474
KNN-CF	0.099	0.106	0.109	0.111	0.113		0.392	0.421	0.440	0.451	0.455
MostPopular	0.006	0.034	0.135	0.159	0.177		0.110	0.148	0.182	0.232	0.267
XGBoost-Hybrid	0.038	0.044	0.049	0.055	0.059		0.304	0.336	0.364	0.389	0.401

Significance — per-user NDCG & set-level Overall Score



Lighter = better. Violins = per-user NDCG distribution; bars = set-level Overall Score ± bootstrap std; * = significant.

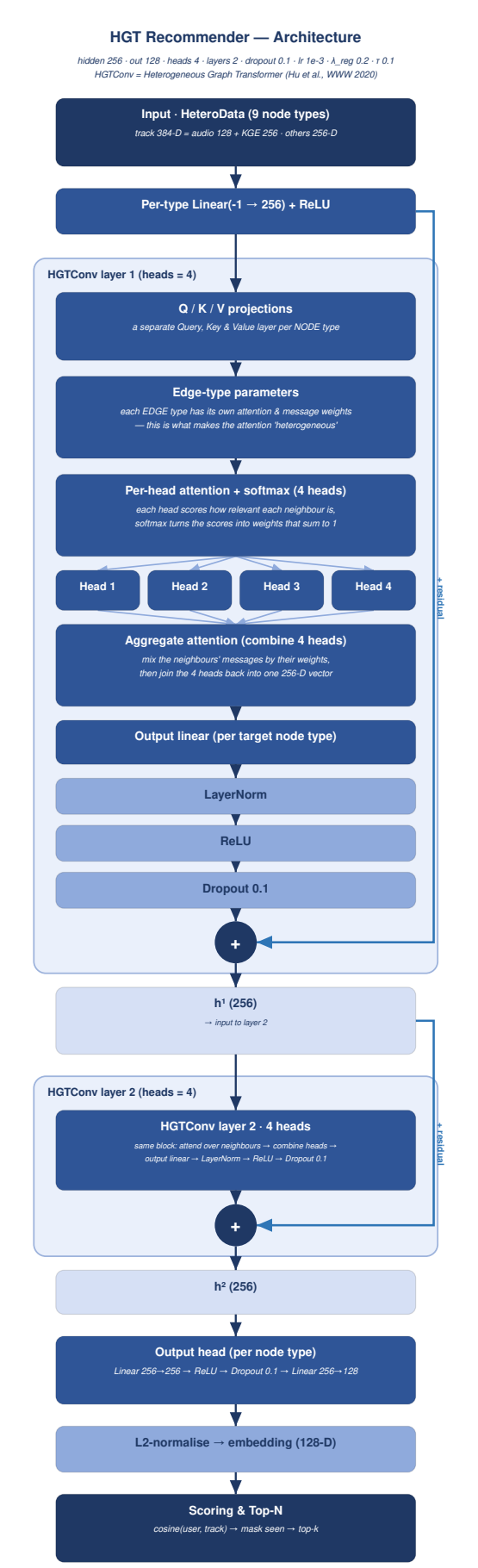
- HGT trails KNN-CF for short lists but **overtakes it** for $K \geq 50$: Overall Score **0.474 vs 0.455** and NDCG **0.148 vs 0.112** at $K=100$, with the **lowest popularity bias (0.090)** throughout.
- MostPopular's high raw NDCG (0.114) is a **popularity artifact**: near-zero coverage (0.006), extreme bias (0.611) → composite 0.148.

References

[1] Z. Hu et al. Heterogeneous Graph Transformer. *WWW*, 2020.
[2] Z. Sun et al. RotatE: KG embedding by relational rotation. *ICLR*, 2019.
[3] C. Raffel. Learning-based methods for comparing sequences (Lakh MIDI). PhD thesis, 2016.
[4] T. Bertin-Mahieux et al. The Million Song Dataset. *ISMIR*, 2011.
[5] The Echo Nest. Taste Profile subset of the MSD, 2011.
[6] C. McKay et al. jSymbolic 2.2. *ISMIR*, 2018.
[7] Y. Raimond et al. The Music Ontology. *ISMIR*, 2007.
[8] H. Wang et al. KG Convolutional Networks for Recommender Systems. *WWW*, 2019.

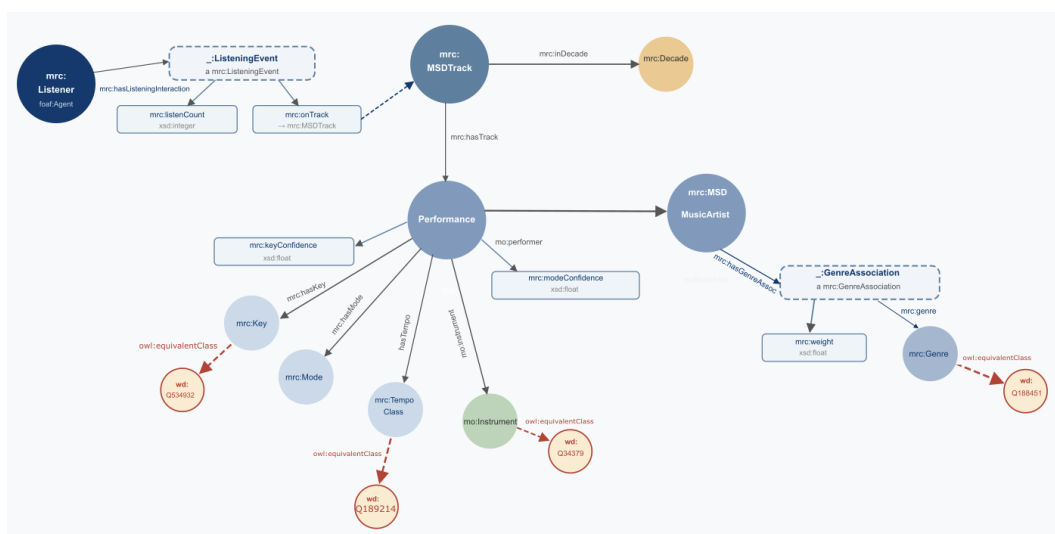
[9] X. Wang et al. KGAT: KG Attention Network for Recommendation. *KDD*, 2019.
[10] X. Liu et al. Music recommendation via KG & multi-task learning. *Sci. Reports*, 2024.
[11] S. Rendle et al. BPR: Bayesian Personalized Ranking. *UAI*, 2009.

HGT Layer



HGTConv (exploded): per-edge-type $K/Q/V$, typed attention, aggregation.

Knowledge Graph



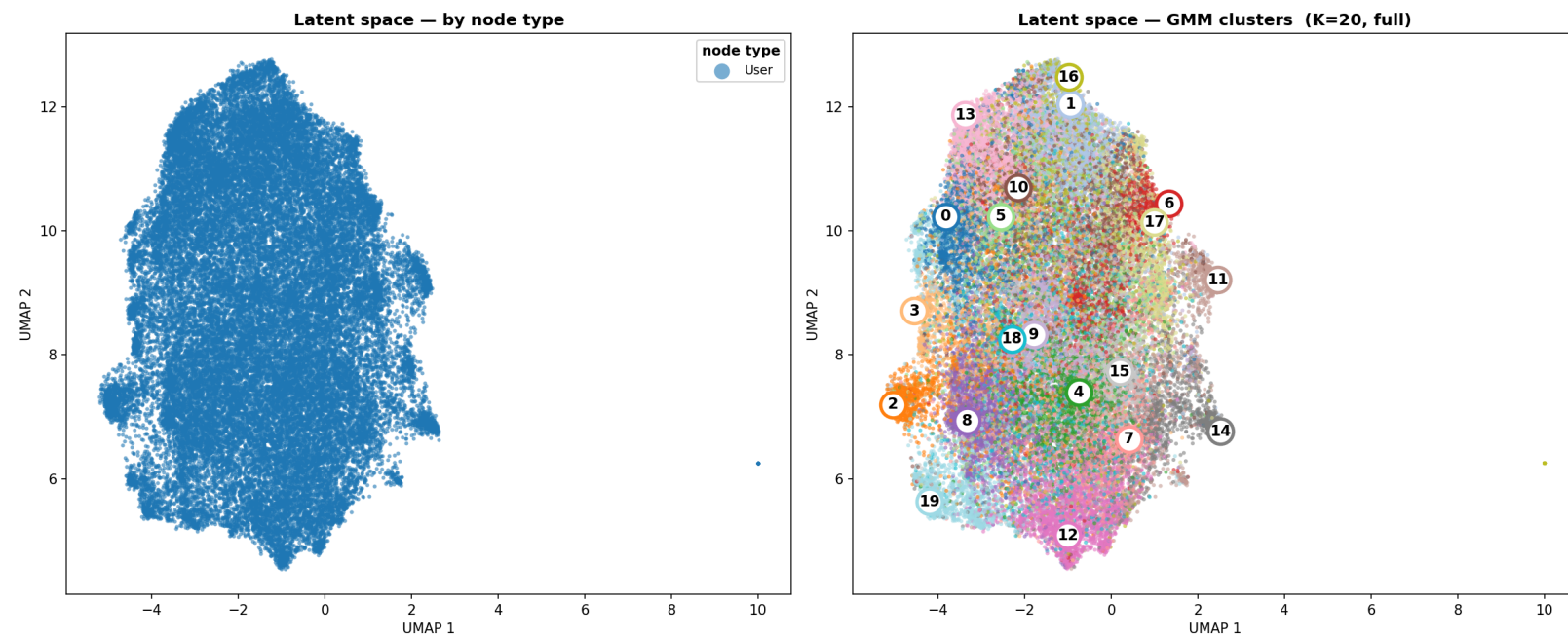
TBox: ontology classes, properties and Wikidata alignments [7].

Knowledge Graph — Entity Composition	
12.8 million triples total	
Entity Type	Count
Listeners	285,037
Tracks	7,013
Performances	7,013
Artists	3,668
Genres	645
Instruments	129
Keys	12
Tempo Classes	9
Decades	7

Entity composition.

- 286,889** users [5], **7,050** songs (Lakh MIDI [3] ↔ MSD [4], DTW ≥ 0.55); songs carry 1,525 symbolic descriptors [6] + audio metadata.
- Genres / tempo / mode linked to **Wikidata** (`owl:equivalentClass, sameAs`); reified blank nodes encode weighted listening & genre events.

Latent Space

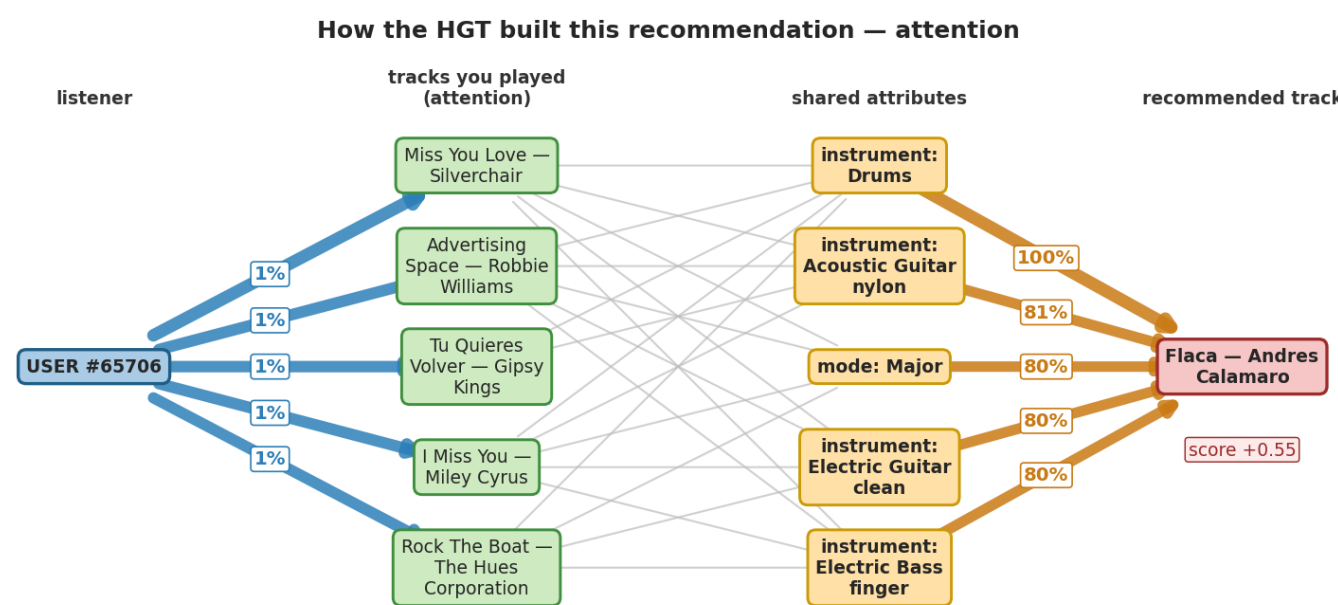


UMAP of the HGT user embeddings; BIC-selected GMM clusters (numbered centroid badges).

- A Gaussian Mixture over the learned user vectors yields interpretable **listener archetypes** (top genres / tempo / mode / decade).
- These archetypes drive the **cold-start** persona app: a new user is mapped to the nearest archetype.

Explainability

Same rec, 3 rationales - USER #65706 -> Flaca — Andres Calamaro (cosine +0.55)



Left edge = each track's attention share (% of your incoming attention; top 5 of 134 (5% of its attention)). Right edge = support = the share of that attention sitting on tracks that carry the attribute. Softmax attention, averaged over heads.

Faithful attention traces a rec to the user's tracks → shared attributes → recommendation.

- Captured HGT **attention** (not a post-hoc proxy) explains *why* each track was recommended, attribute by attribute.
- Cross-checked against **occlusion Δ score** and **Integrated Gradients** — the three agree on the top drivers.

Conclusions & Future Work

- The HGT **matches and overtakes** the strongest CF baseline at depth ($K \geq 50$) on NDCG, coverage & Overall Score, with the **lowest popularity bias** of all recommenders.
- Symbolic + audio content in a **typed KG** gives interpretable, diverse recommendations; brand-new users are served through **listener archetypes** (cold-start).
- Captured HGT **attention** yields faithful, attribute-level explanations, corroborated by occlusion Δ score and Integrated Gradients.

Future work

- Adaptive popularity control**: learn / anneal the debiasing weight λ and temperature τ to trade relevance against fairness on demand.
- A **hybrid HGT + CF** re-ranker that exploits the depth crossover — CF for short lists, the HGT for long, diverse ones.
- Richer, multi-hop KG signals** (lyrics, mood, social ties) and deeper propagation beyond the single user→track hop.
- Temporal / session-aware** modelling, plus a human study of how useful the attention explanations are to listeners.